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Assessment of the Quality of Soils of Coconut Cultivating Lands via Geographical Information Technology Approaches: A Case Study in Bandirippuwa Coconut Estate, Lunuwila

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ABSTRACT

Coconut (*Cocos nucifera* L.) is one of the major plantation crops in Sri Lanka which contributes significantly to ensure food and nutritional security to a larger segment of the population. Maintaining optimum levels of chemical and physical soil properties is a fundamental requirement for better palm growth and higher nut yield. Soil chemical and physical properties act as indicators of soil quality. For this study, geostatistical techniques were used which applies statistical methods in spatial interpolation for predicting spatial variability of soil properties. This study was carried out in coconut cultivating land of Bandirippuwa, to assess the quality of soils of coconut cultivating lands via Geographic Information System (GIS). From that, the soil quality was determined and enabling in deciding whether the selected land areas are within the ideal values or not for further continuation of coconut cultivation and in need for future improvements. Therefore, some selected physical and chemical properties were determined. Soil pH, EC (electrical conductivity), SOC (soil organic carbon) were the tested chemical properties while soil texture and bulk density (BD) were the physical properties determined during this research. During this study, soil quality was assessed via raster processing technique of raster calculator using Arc GIS 10.2 software. The study concludes that in the selected land plot, coconut near the water source and cattle grazing areas were the excellent areas in soil quality while mono-cropping area was having the moderately good and poor soil quality and the intercropping and steepy area showed both good and moderately good soil quality conditions in Bandirippuwa coconut estate.

KEYWORDS: Coconut cultivation, Raster processing techniques, Soil chemical properties, Soil quality

INTRODUCTION

Coconut (*Cocos nucifera* L.) is a perennial crop having multiple uses to the humans. Therefore, it is called as "*Kapruka*". Sri Lanka is the fifth largest coconut producer in the world (Anon, 2018b). The main coconut growing districts in Sri Lanka, referred to as the "coconut triangle", is comprised of Kurunegala, Puttalam and Gampaha districts. Coconut is widely distributed in sandy loam type of soil in the Intermediate zone of Sri Lanka. Bandirippuwa estate is one of the coconut estates managed by the Coconut Research Institute of Sri Lanka and located in the Intermediate zone (IL1a). The dominant soils of IL1a Agro-Ecological Zone are the Red-Yellow Podosolic soils with mottled sub soils together with regosols on old red and yellow sands (Panabokke, 1996).

Soil chemical and physical properties act as an indicators of soil quality. Furthermore, chemical, physical and biological properties are essential for better crop growth and crop yield. Soil is the basis for any agricultural or natural plant communities. It is a living, dynamic ecosystem that provides the natural medium for the growth of plants. Healthy soil is a fundamental resource for high coconut yield. The soil quality of coconut lands are decreasing due to high application of inorganic fertilizers irrespective of the recommended levels and other chemicals leading to changes its ideal soil conditions for coconut palm growth.

In order to enhance the coconut yield and soil quality in coconut lands, it is necessary to follow correct fertilizer recommendations at correct time. Maintaining the soil quality *via* continuous supplementation of essential nutrients is a fundamental requirement for higher coconut production. (Samarasekara *et al.*, 2017). Therefore, to facilitate the wellbeing of the coconut palms, it is vital to minimize negative soil impacts and maintenance of sustainable soil quality conditions in coconut cultivated lands.

In general, there exist a big variation in soil properties of a selected small coconut land area. Therefore, geostatistical approaches are used to study on heterogeneous soils. Using geostatistical techniques it enables to quantify the spatial variability of soils. Furthermore, it facilitates to map the given soil properties.

(Eranga *et al.*, 2015). In addition, soil properties vary spatially from a small field to a larger regional scale affected by both soil forming factors and soil management practices (Cambardella and Karlen, 1999). The variation of soil properties should be monitored and quantified in order to understand the effects of land use management systems on soils. As the result, geostatistical methods have been developed and used successfully for predicting spatial variability of soil properties. After introduction of Digital Soil Mapping concept (DSM), mapping of soil properties has become more popular (McBratney *et al*., 2003).

The general objective of this study was to assess the quality of soils and its variation of a coconut cultivating land using simple GIS approach. Specific objectives of the study were to develop continuous surface maps at higher spatial resolution (10 m) for identified soil properties and to use raster based operations to derive a simple soil quality index.

MATERIALS AND METHODS

Study Site

This experiment was conducted in the Bandirippuwa coconut estate, Bandirippuwa, Lunuwila (7°19' 39" N, 79° 52' 5" E) located in the Intermediate zone of Sri Lanka.

Sampling Plan

Total 63 ha of land from different three land use types *viz.* (intercropping and steepy area, mono-cropping area, coconut near water source and cattle grazing area) were demarcated using Trimble JUNO SB GPS receiver (Trimble Navigation Limited, USA). Soil samples were collected using a stratified random sampling design based on the land management zones. Total numbers of 92 soil samples were collected from 46 sample points from 0-0.15 m and 0.15-0.30 m depths. Sampling locations were allocated for each strata proportionately based on the area.

Laboratory Soil Analysis

The laboratory analysis was carried out in Soils and Plant Nutrition Division of Coconut Research Institute, Lunuwila, Sri Lanka and at the Faculty of Agriculture and Plantation Management, Wayamba University of Sri Lanka, Makandura, Gonawila (NWP) from June to October 2018.

Samples were air dried under room temperature and sieved with a 2mm sieve prior to the analysis. The pH and EC of soil samples were determined by the 1:5 soil: water suspension method (Black, 1965). The pH and Electrical Conductivity (EC) values were measured electrometrically using Orion Star A215 pH and EC meter and Soil Organic Carbon (SOC) percentage was measured by Walkley and Black method (Walkley and Black, 1934). Soil bulk density (BD) was determined using core sampling method (Blake and Hartge, 1986), while the soil texture was determined using hydrometer method (Dharmakeerthi *et al*., 2007).

Spatial Modelling of the Continuous Soil Properties

Geostatistical models were fitted for all the continuous soil properties considering two depth intervals. Then ordinary kriging was performed to estimate the spatial variability of the considered soil properties across the study area. Mapping was carried out at 10 m spatial resolution**.**

Assessment of Soil Quality

The most suitable ranges of each soil property for the coconut were utilized to reclassify the developed continuous surface maps with the aid of ordinary kriging. The best range was ranked as one (1).

Allocation of Ranks for Classified Digital Layers

The numerical ranking was done for each chemical and physical properties by considering the most suitable ranges for coconut cultivation. The ideal pH range for the coconut cultivation is 5.5 to 7.5. According to the classification 5.5 - 6.1 was categorized as rank 1.

The ideal EC range for the coconut is below 4000 µS/cm. The favorable EC range for the top soil was 22.03 – 27.71 µS/cm and for the sub soil was 11.67 – 84.86 µS/cm which were categorized as rank 1.

The ideal range of SOC for coconut cultivation is below 2 %. According to the classification SOC was ranked under 5. The ideal range for the sand in coconut cultivation is above 60 %. According to the classification, sand content of 70 – 85 % was categorized under rank 1.

For the coconut cultivation, the ideal range of silt is below 30 %. According to the classification if the silt content is less than 11 %, it was classified as rank 1 while if it is more than that, it was classified as rank 2. Below 10 % is the ideal range for clay. Hence it was ranked under 1.

The ideal range of bulk density for coconut cultivation is 1.4–1.6 g/ cm^3, which was ranked under 1. If the pH, sand, clay and bulk density are away from the ideal ranges, comparative values were assigned for them (Table 1).

Table 1. Allocation of ranks for classified digital layers

Parameter	Ideal Ranges	Classification	Rank
pH	5.5 - 7.5	5.5 – 6.1	1
		4.5 – 5.4	3
EC	0 – 4000 µS/ cm	11.67 – 84.86 µS/ cm (Sub Soil)	1
		22.03 – 27.71 µS/ cm (Top Soil)	1
SOC	< 2%	< 1%	5
Sand	> 60 %	70 – 85 %	1
		85 -95 %	2
		> 95 %	5
Silt	< 30 %	< 11	1
			2
Clay	< 10 %	< 10%	1
		10 – 15 %	2
		> 15 %	3
BD	1.4 – 1.6 g/ cm3	1.4 – 1.6	1
		1.2 – 1.4	2
		< 1.2	3

Note- Rank 1 was assigned to the best range, EC- Electrical Conductivity, SOC- Soil Organic Carbon, BD- Bulk Density

Soil chemical status, soil texture, soil physical status and overall soil status were calculated for both top and sub soil. Final soil quality was calculated using both physical and chemical status. Following equations were used to analyze the soil quality by employing the raster processing techniques of raster calculator.

Soil chemical status for each depth = pH × EC× SOC

Soil texture for each depth = Sand × Silt × Clay

Soil physical status for each depth = Soil BD × Soil texture

Soil status for each depth = Physical status × Chemical status

Final soil quality = Top soil status × Sub soil status

Where, EC is the Electrical Conductivity (µS/cm), SOC (%) is the Soil Organic Carbon percentage and BD is the Bulk Density (g/cm^3).

Finally, the quality of the soil was decided based on the numerical values resulted from the raster manipulation operation in GIS. Then the land extent under each category was calculated in GIS (Table 2).

Table 2. Land area under each category

Category	Area (ha)
Excellent	7.94
Good	17.46
Moderately Good	24.31
Poor	12.95

The areas under each category defined in Table 2, was further categorized according to land use types as in Table 3.

Table 3. Land area Vs. land use type

Land Use Type	Category	Area (ha)
Coconut near water sources and cattle grazing	Excellent	6.43
	Good	3.92
	Moderately Good	0.02
Mono-cropping	Moderately Good	7.82
	Poor	11.2
Intercropping and Steep	Excellent	1.49
	Good	13.54
	Moderately Good	16.46
	Poor	1.74

RESULTS AND DISCUSSION

Descriptive Statistics

The pH of all the sampling points of top layer (0 - 0.15 m) ranged from 4.15 to 6.29 (Table 4). According to Dharmakeerthi *et al*., (2007) that range is known as very strongly acidic to slightly acidic. Furthermore the pH of all the sampling points of sub layer (0.15 - 0.30 m) ranged from 4.12 to 6.15. Therefore it can be categorized into the range of very strongly acidic to slightly acidic (Dharmakeerthi *et al*., 2007). The ideal pH for coconut cultivation is 5.5 - 7.5 (Liyanage, 1999). Both top and sub layers were ranged within the ideal pH level.

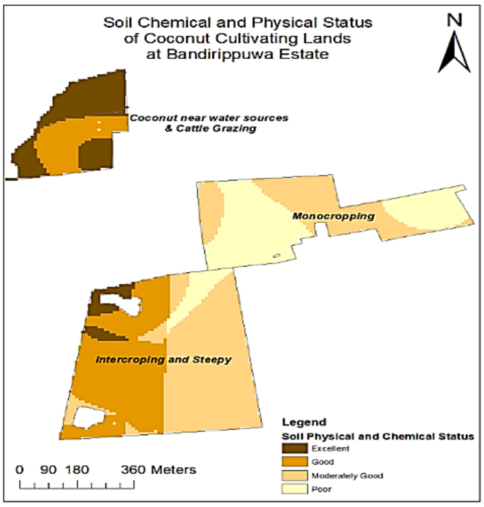

Figure 1. Soil chemical and physical status of coconut cultivating lands at Bandirippuwa estate

The value of EC in the top layer (0 - 0.15 m) was varied between 12.81 - 56.87 µS/cm (Table 4). EC in the sub layer was varied between 12.45 - 84.27 µS/cm. As per the handbook of soil survey and agricultural and evaluation in the tropical and sub tropics (Landon, 1991), it revealed that the ideal EC level for coconut cultivation is 0 - 4000 µS/cm. Both top and sub layers were ranged within the ideal EC level.

Table 4. Descriptive statistics of pH, EC, SOC of top layer and sub layer

Parameters	Depth (m)	Minimum	Maximum	Mean	Standard Deviation	1st Quartile	Median	3rd Quartile
pH	0.00-0.15	4.15	6.29	5.43	0.53	4.99	5.51	5.86
	0.15-0.30	4.12	6.15	5.11	0.58	4.61	5.12	5.59
EC (μS/cm)	0.00-0.15	12.81	56.87	24.72	9.67	17.22	21.15	28.76
	0.15-0.30	12.45	84.27	28.09	13.94	18.04	24.72	34.83
SOC (%)	0.00-0.15	0.05	0.8	0.46	0.19	0.3	0.48	0.62
	0.15-0.30	0.12	0.55	0.37	0.15	0.41	0.21	0.51

Depth of top layer=0.00-0.15 m; depth of sub layer=0.15-0.30 m; SOC= Soil Organic Carbon; EC= Electrical Conductivity

SOC content in the top layer of soil was varied within the ranged of 0.05 - 0.8 % (Table 4). It was varied between 0.12 - 0.55 % in the sub layer. The most suitable range of SOC falls in between 2 - 20 % (Metson, 1957). The obtained values of SOC for both top and sub layers were below than the most suitable range.

The resulted values from the raster manipulation operation in GIS were categorized as Excellent, Good, Moderately Good and Poor (Table 2). Three kinds of land use types were identified in the study site, namely Monocropping, Intercropping and steepy area, Coconut near water sources and cattle grazing (Figure 1). Out of the total extent (63 ha), 7.94 ha of land area was categorized as "Excellent", while 17.46 ha of land area was categorized under "Good". From the remaining, 24.31 ha of area was categorized as "Moderately Good" and the rest (12.95 ha) was categorized under "Poor" (Table 2).

In this study, the coconut land near to the water sources and cattle grazing area, (6.43 ha of area) was categorized under "Excellent", while 3.92 ha of area was categorized under "Good" and 0.02 ha of area was categorized under "Moderately Good". In mono-Cropping area, 7.82 ha was categorized as "Moderately Good" and 11.20 ha of area was categorized as "Poor". In intercropping and steepy area, 1.49 ha of area was categorized as "Excellent", while 13.54 ha, 16.46 ha and 1.74 ha of areas were categorized as "Good", and "Moderately Good" and "Poor" respectively (Table 3).

According to the final result of raster based grid operations, the resulted numerical value for the mono-cropping area was categorized as "Moderately Good and Poor" (Figure 1). During the study, the soil samples were collected from center square of the coconut land except the manure circle. Therefore, there would be a lack of nutrients in those areas. Areas which soil samples were collected were partially covered the ground which has directly exposed to the sunlight due to lack of cover crops, especially when the palms are young. Therefore the moisture would be evaporated while organic matters get turn over rapidly (Liyanage *et al*., 1986). Due to the cultivation of a single crop for a period of time continuously, most of the chemical and physical properties get degraded resulting reduction in soil quality.

Intercropping and steepy area was categorized as "Good and Moderately Good" (Figure 1). This could be due to the fact that intercropping coconut lands with other crops improve soil fertility by restructuring organic matter and with altered micro environment for early bearing of coconut while reducing soil erosion and the temperature (Anon, 2018a). Coconut do not add much organic matter to the soil to improve fertility unlike other perennial crops. Thus, practicing intercropping would facilitate the improvement of soil quality on behalf of the addition of crop litter to the soil (Liyanage *et al.,* 1984). Intercropping also gives a greater stability of yield over mono-cropping. Compared to mono-cropping, intercropping provides year round ground cover for long time period and protect the soil from soil erosion (Gebru, 2015).

Coconut near water sources and cattle grazing area was categorized as "Excellent" and some of the parts were categorized as "Good" (Figure 1). Cattle grazing under coconut cultivation enables recycling of nutrients through the accumulations of urine and dung as an organic fertilizer. It enhances the organic matter content in the soil thereby enriching the coconut cultivation by reducing the cost for chemical fertilizer. Cattle grazing will eradicate palatable weed species from the coconut land. It also reduces the cost for weeding in coconut cultivation (Liyanage *et al*., 1993). Optimum moisture content is required for better growth of coconut palm. Optimum soil moisture may also facilitate the efficient nutrient uptake in moist soil. Additionally, optimum moisture level enhance the microbial activities near the root surface of coconut. The increased soil fertility was observed due to the richness in physical and chemical properties of that soil than the other selected areas. There was no any competition between grasses and coconuts for moisture and nutrients. Grasses act as cover crops and green manure for coconut cultivation. In addition, they become fertilizer after their life cycle by enriching the soil nutrients. High fertility status of the supporting soils is necessary for high productivity of coconut lands.

CONCLUSIONS

This study demonstrated how geospatial technologies can be coupled with point based soil observations to create soil quality assessment for a coconut cultivating land across the landscape. Grid-based analysis revealed that out of total land extent in the Bandiruppuwa, only 7.94 ha of land is at the optimum condition. Developed soil quality map can be used in future planning of land management practices to enhance the soil quality and obtain maximum economical returns through enhanced land use efficiency.

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